



HW1

Ternary Raster Operations (ROP3) for Computer Graphics

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Introduction

- Raster Operation (ROP)
 - Provides bit-level functions for computer graphics e.g. on PCs and mobile phones
 - Consist of total 256 different functions
 - Input
 - $[4*N-1:0]$ din (Candidates for **Pattern, Source, Destination**)
 - din_valid (Valid data indicator)
 - $[7:0]$ mode (Function mode)
 - $[4:0]$ sel (Selection for candidates)
 - Output
 - $[N-1:0]$ result
 - dout_valid

Selection

- Split Din into four N-bits slice

$$\begin{aligned} W &= \text{din}[4 * N - 1 : 3 * N] & X &= \text{din}[3 * N - 1 : 2 * N] \\ Y &= \text{din}[2 * N - 1 : 1 * N] & Z &= \text{din}[1 * N - 1 : 0 * N] \end{aligned}$$

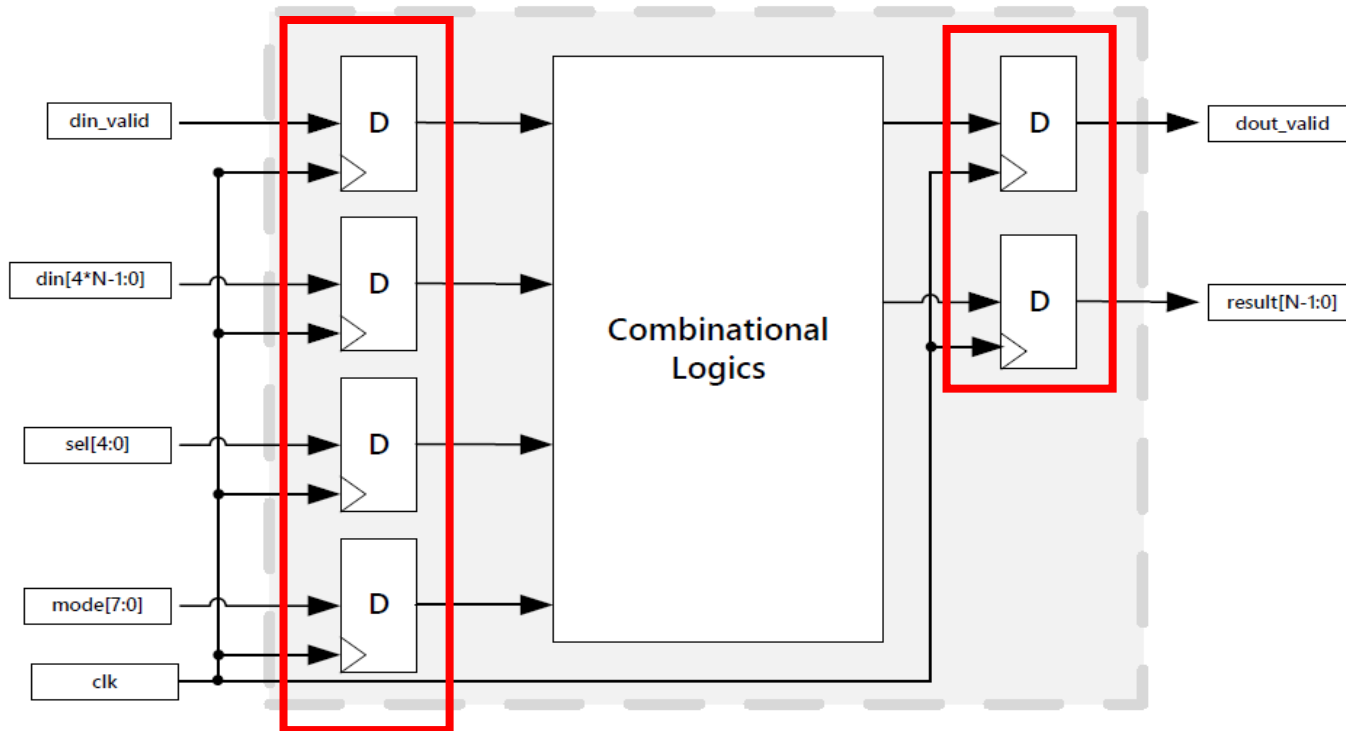
- Select Pattern, Source, Destination from the table

Sel[4:0]	P	S	D
5'd0	W	X	Y
5'd1	W	Y	X
5'd2	X	W	Y
5'd3	Y	W	X
5'd4	X	Y	W
5'd5	Y	X	W
5'd6	W	X	Z
5'd7	W	Z	X
5'd8	X	W	Z
5'd9	Z	W	X
5'd10	X	Z	W
5'd11	Z	X	W

⋮

Schematic view

Add D flip-flops for all IO



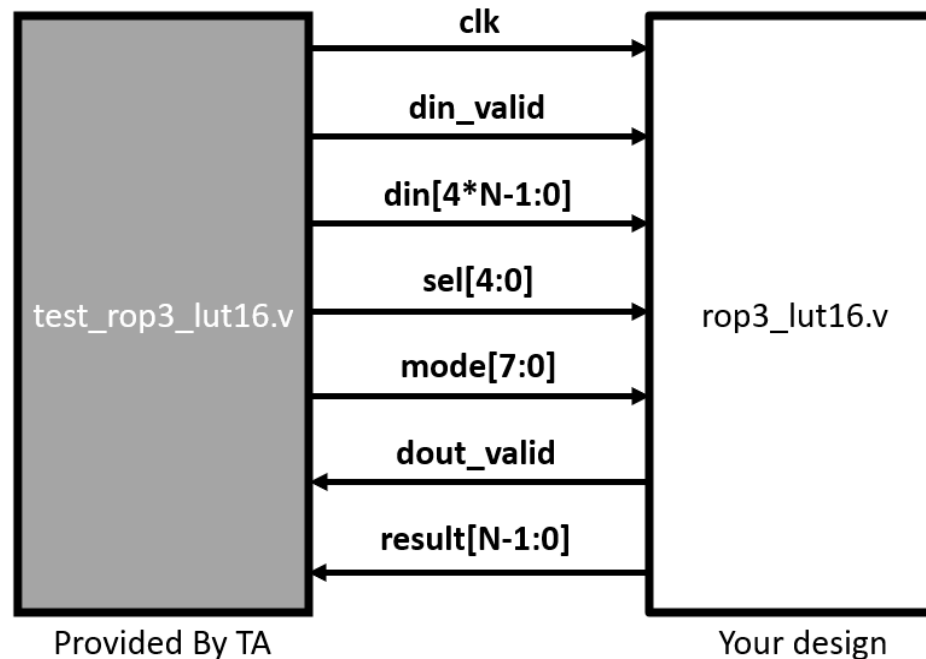
Fifteen ROP3 functions (for Part 1)

Pattern (P)	1 1 1 1 0 0 0 0	Mode (Hex)	Function Name	Boolean Equation
Source (S)	1 1 0 0 1 1 0 0			
Background (D)	1 0 1 0 1 0 1 0			
Result	0 0 0 0 0 0 0 0	8'h00	BLACKNESS	0
	0 0 0 1 0 0 0 1	8'h11	NOTSRCERASE	$\sim(D S)$
	0 0 1 1 0 0 1 1	8'h33	NOTSRCCOPY	$\sim S$
	0 1 0 0 0 1 0 0	8'h44	SRCERASE	$S \& \sim D$
	0 1 0 1 0 1 0 1	8'h55	DSTINVERT	$\sim D$
	0 1 0 1 1 0 1 0	8'h5A	PATINVERT	$D \wedge P$
	0 1 1 0 0 1 1 0	8'h66	SRCINVERT	$D \wedge S$
	1 0 0 0 1 0 0 0	8'h88	SRCAND	$D \& S$
	1 0 1 1 1 0 1 1	8'hBB	MERGEPAINT	$D \sim S$
	1 1 0 0 0 0 0 0	8'hC0	MERGECOPY	$P \& S$
	1 1 0 0 1 1 0 0	8'hCC	SRCCOPY	S
	1 1 1 0 1 1 1 0	8'hEE	SRCPAINT	$D S$
	1 1 1 1 0 0 0 0	8'hF0	PARTCOPY	P
	1 1 1 1 1 0 1 1	8'hFB	PATPAINT	$D P \sim S$
	1 1 1 1 1 1 1 1	8'hFF	WHITENESS	1



Part 1

- Implement the fifteen ROP3 functions in Table 1
- Complete rop3_lut16.v
- Test your RTL by the provided testbench



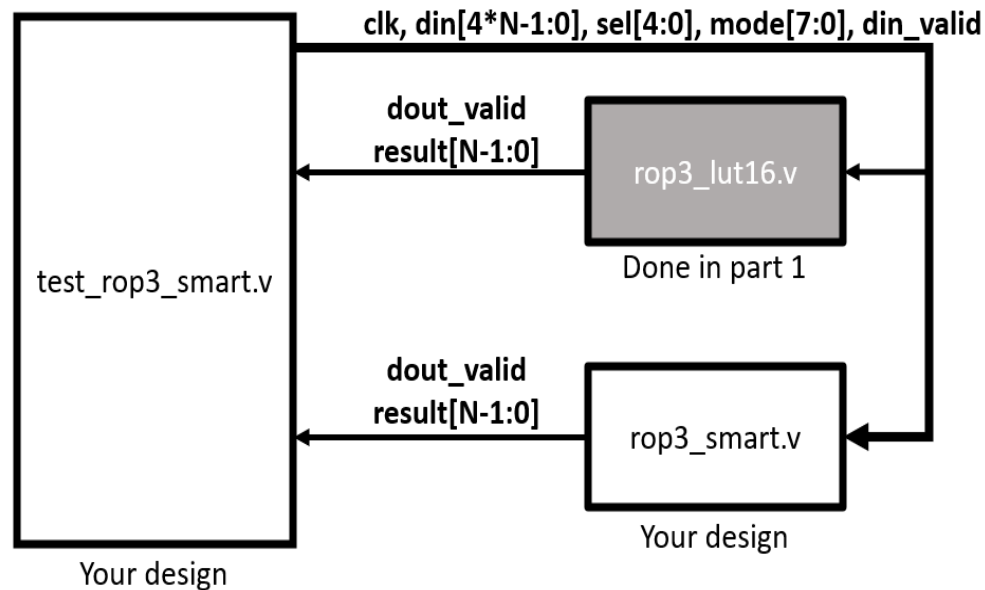
Smart formulation of ROP3 (for Part 2)

- Works for each bit position i independently

```
temp1 [7:0] = 8'h1 << {P[i], S[i], D[i]};  
temp2 [7:0] = temp1 [7:0] & mode [7:0];  
result [i] = | temp2 [7:0];
```

Part 2

- Implement ROP3 by the smart formulation
- Complete rop3_smart.v
- Write a testbench test_rop3_smart.v to test your RTL

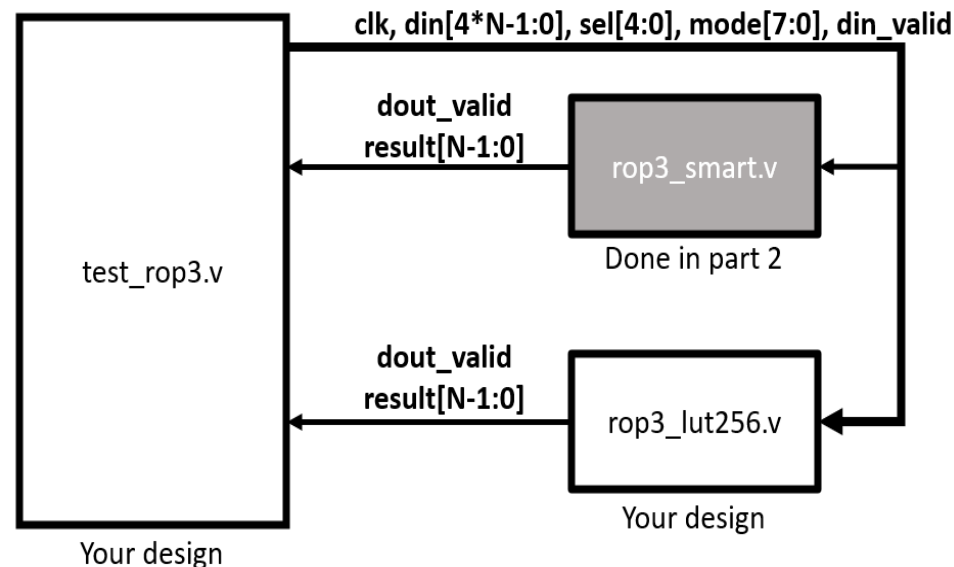


Part 2 Cont'd

- Your testbench should meet some requirements
 - 1) Send identical inputs to rop3_lut16 & rop3_smart
 - 2) Generate all 15 modes, 24 selections and all possible combinations of din
 - 3) Use ``define N` in your testbench to make the bit-length of {din, result} parameterizable
 - 4) Make sure your RTL design and testbench pass under different bit-length N, where $N \in \{4, 8\}$

Part 3

- Find out all 256 Boolean functions for ROP3, and use a large multiplexer to implement it.
- Complete rop3_lut256.v
- Write a testbench test_rop3_lut256.v to test your RTL



Part 3 Cont'd

- Your testbench should meet some requirements
 - 1) Send identical inputs to rop3_lut256 & rop3_smart
 - 2) Generate all 256 modes, 24 selections and all possible combinations of din
 - 3) Use ``define N` in your testbench to make the bit-length of {din, result} parameterizable
 - 4) Make sure your RTL design and testbench pass under different bit-length N, where $N \in \{4, 8\}$



README.txt & Synthesizable RTL

- Write a README.txt to describe
 - 1) Observation in Table 1 and how you find out all 256 functions for ROP3 in Part 3?
 - 2) How you organize your testbench to test your RTL design in Part 2 & Part 3 ?
- All RTL designs should be synthesizable. Submit Spyglass report for all 3 parts.

Submission

- Deadline
 - 10/2 11:59 p.m.
- Submit to eeclass
 - Make sure the file delivery and organization meet the requirement
 - Wrong file delivery or organization will get 10 points punishment
- If you have any question
 - Feel free to ask on eeclass discussion
(maybe other students also have the same question !)
 - Do not show your code or email your code to TA.
Coding by yourself !
- Using generative AI is not recommended.
 - If used, please explain in the report how it was applied